

Hands-on Exercise: Appreciating Frontend Development Approaches

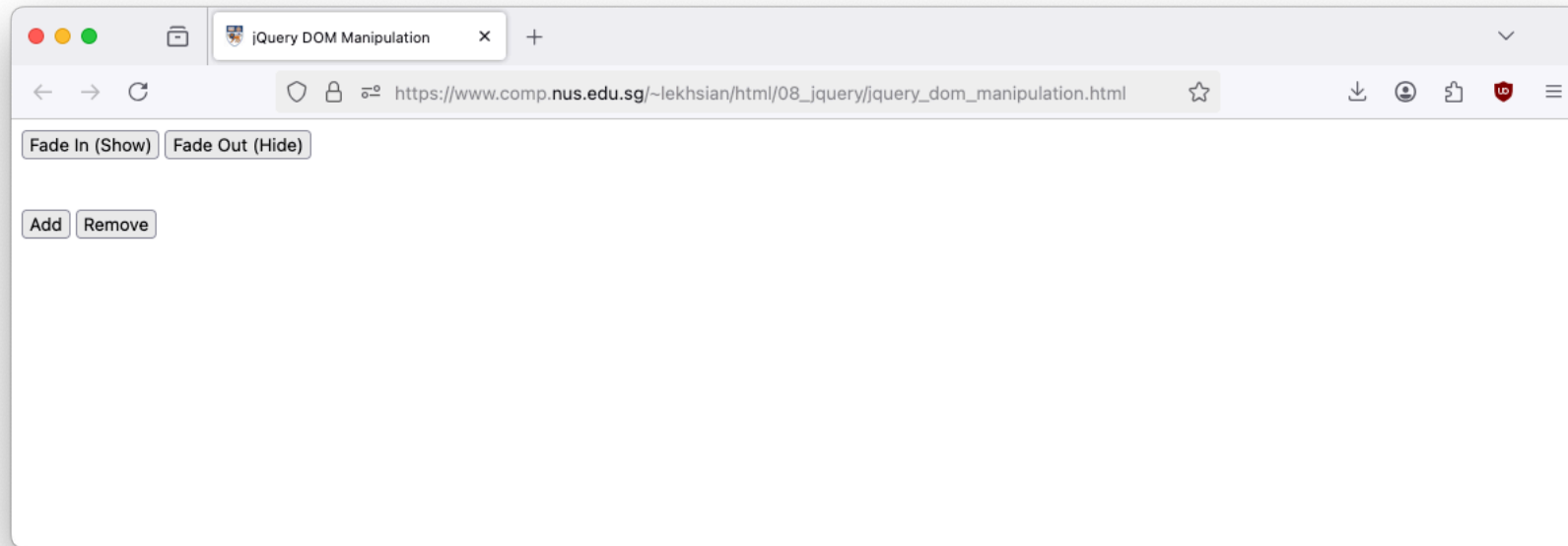
Traditional Approach to Frontend Development

- Developers handle the **Document Object Model (DOM) manipulation**
 - Developers are in charge of making changes to the DOM when there are changes in the data 

Task 1: DOM Manipulation

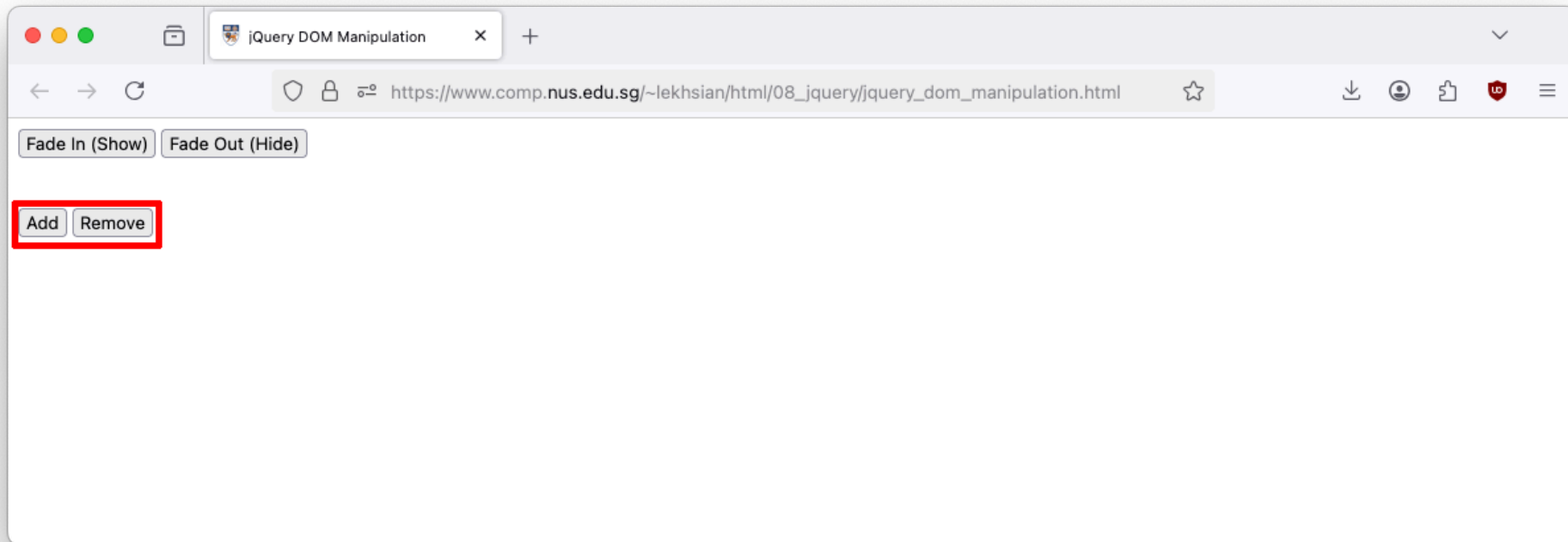
1. Visit

https://www.comp.nus.edu.sg/~lekhsian/html/08_jquery/jquery_dom_manipulation.html to see what it means that developers are handling DOM Manipulation



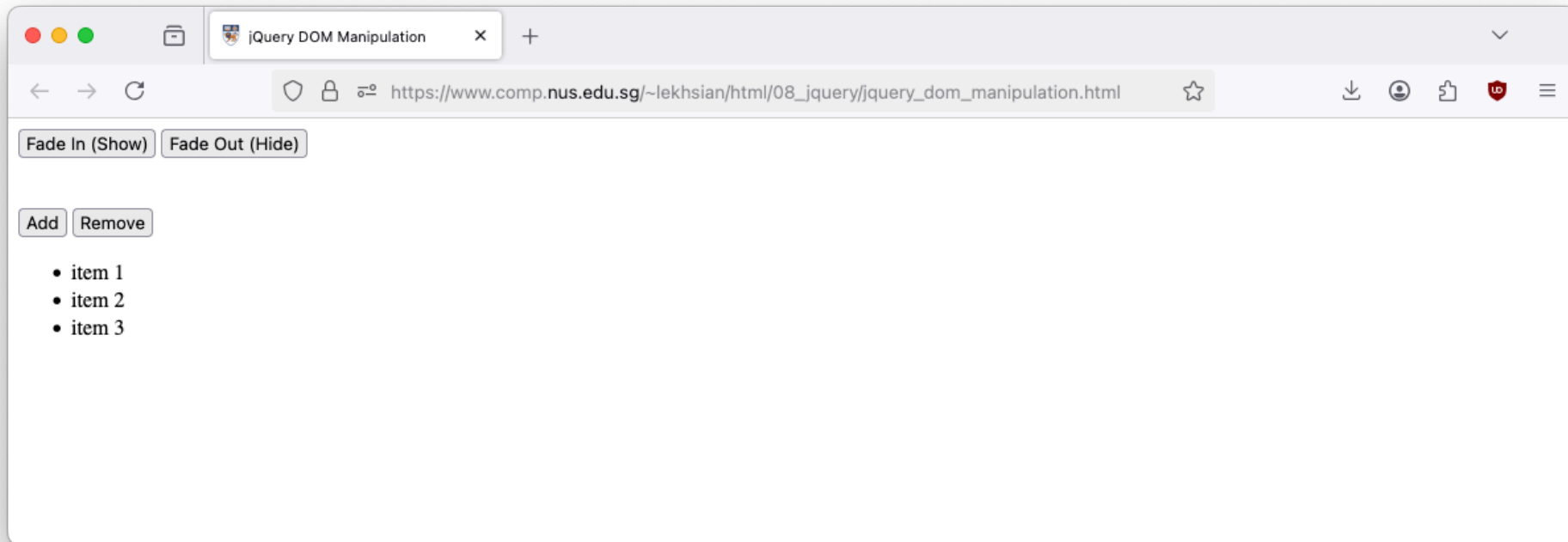
Task 1: DOM Manipulation

2. Press the **Add** or **Remove** buttons



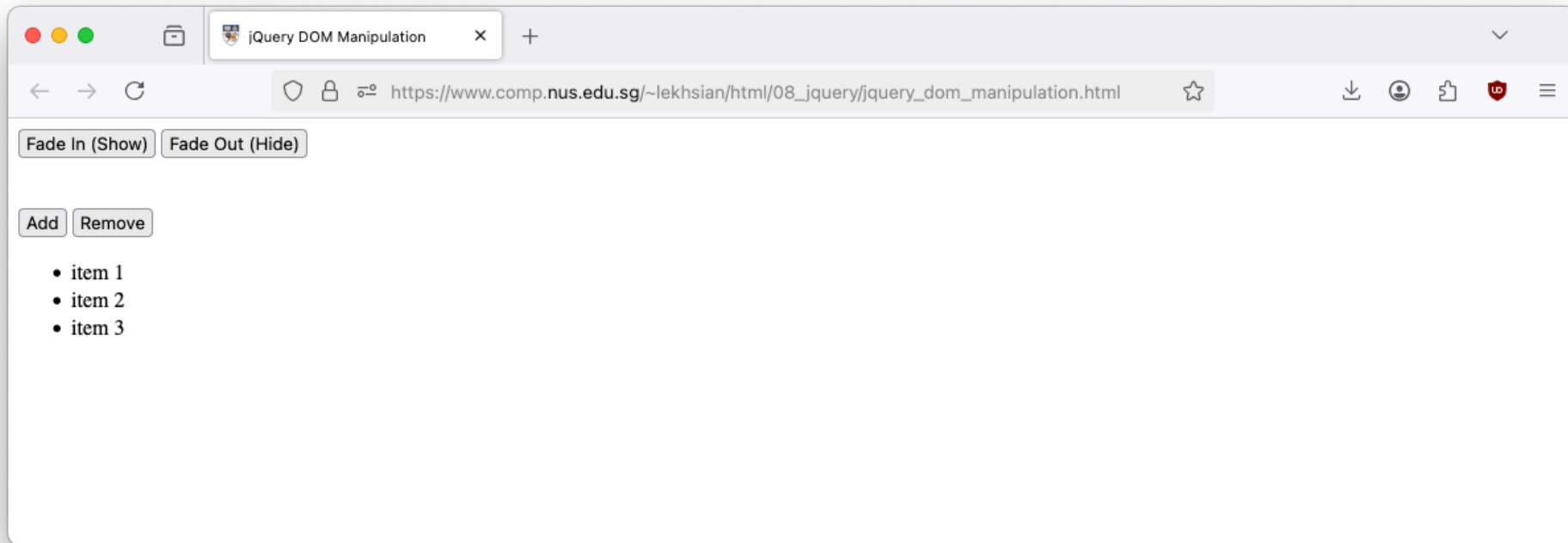
Task 1: DOM Manipulation

3. You will realize that when the **Add** buttons are pressed, items are added to the page



Task 1: DOM Manipulation

4. The DOM in this case is programmatically manipulated by developers once the users press a button, which will trigger a Javascript function call



Task 1: DOM Manipulation

```
<button onclick="addItem()">Add</button>  
<button onclick="removeItem()">Remove</button>  
<ul id="list"></ul>
```

Add Remove

https://www.comp.nus.edu.sg/~lekhsian/html/08_jquery/jquery_dom_manipulation.html

Task 1: DOM Manipulation

```
const listElement = $("#list");
let count = 0;

function addItem() {
  listElement.append("<li>item " + ++count + "</li>");
}

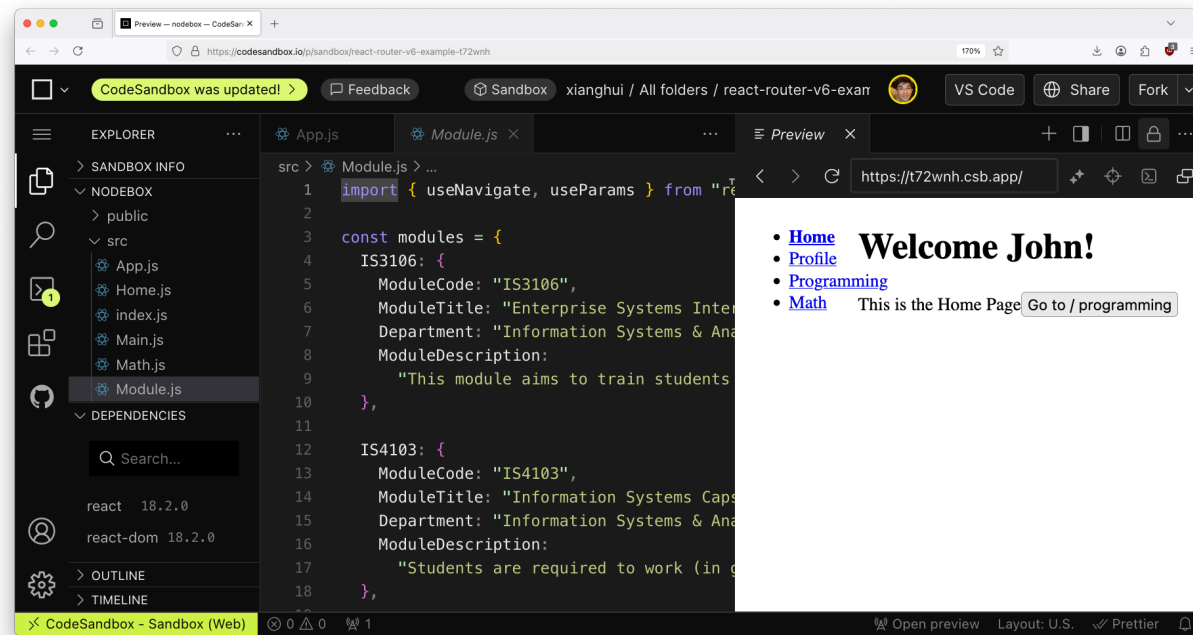
function removeItem() {
  if (count >= 1) {
    listElement.children().last().remove();
    count--;
  }
}
```

https://www.comp.nus.edu.sg/~lekhsian/html/08_jquery/jquery_dom_manipulation.html

Task 2: Single Page Application

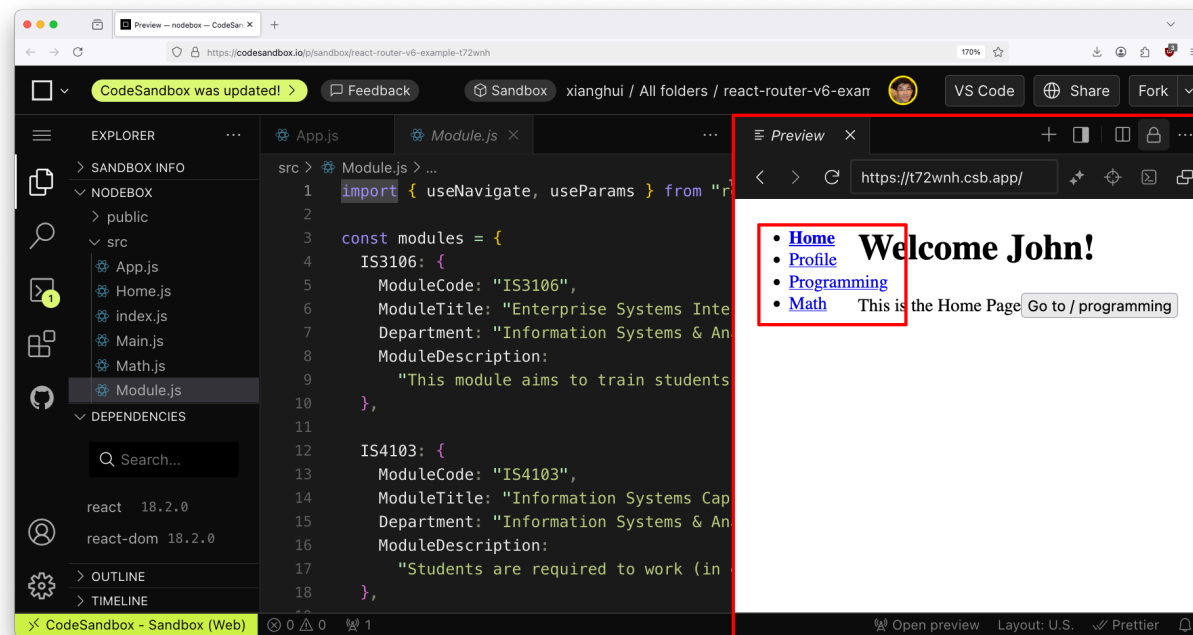
- To better appreciate how Single Page Application works

1. Visit <https://codesandbox.io/p/sandbox/react-router-v6-example-t72wnh>



Task 2: Single Page Application

2. You do not need to understand the codes for now, try pressing on any of the links on the **Preview** pane



Task 2: Single Page Application

3. You will realize that the URL changes
4. There is only one physical page, but this view is programmatically modified by the frontend framework to give the users the impression of multiple pages!

